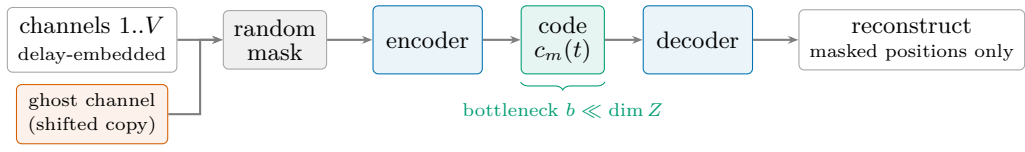
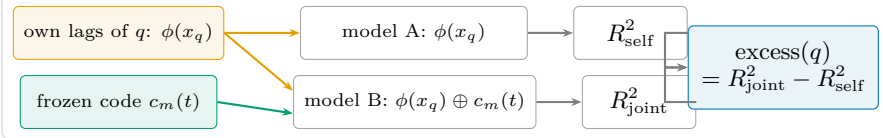


(a) Train M masked autoencoders (unsupervised)



(b) Readout per variable q (ridge regression; no retraining)



(c) Reading the score

excess > 0 : the system's state predicts q 's next value beyond q 's own past $\Rightarrow q$ is *driven*.

excess ≈ 0 : q 's own history already suffices $\Rightarrow q$ is *autonomous*, or a pure *source* (invisible by design).

ghost $\neq 0$: the analysis is producing artifacts; the scan is invalid and must be discarded, not thresholded.